Optical Thin-Film Theory

Macleod's matrix formalism and numerical modelling

MASTER'S LEVEL (M2)

Online version and interactive simulator: **tpcouchesminces.vercel.app**

This document sets out the physics and the equations used to compute the optical properties of a thin-film stack. Written for students of numerical physics and optics, it weaves together physical intuition (interference), Maxwell's equations, and the numerical implementation through the transfer-matrix method (TMM).

Level 1	Level 2	Level 3
Intuition & physical concepts	**Formalism & engineering calculations**	**Expertise & critical pitfalls**
Discovering interference, visual analogies and key diagrams.	Maxwell's equations, 2×2 matrices, computing R and T, Python code.	Complex branch cuts, wave profile $\lvert E(z)\rvert^2$, Poynting flux.

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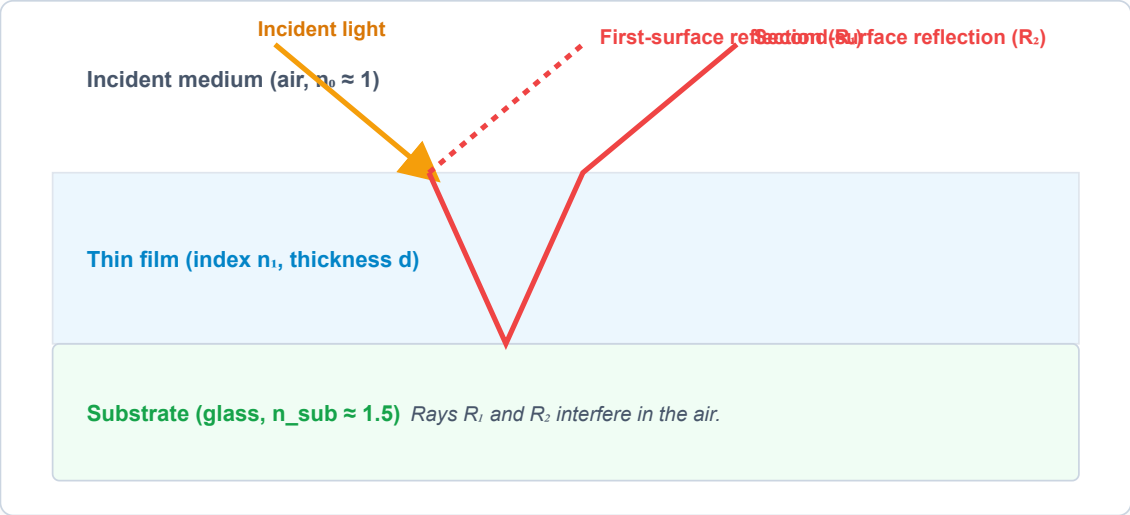
1. Introduction to thin-film interference Level 1

THE PHYSICAL INTUITION

Have you ever noticed the iridescent colours on a soap bubble or an oil slick? That is thin-film optics at work. Part of the light is reflected at the first interface (air/oil), while another part enters the film, propagates, and is reflected at the second interface (oil/water).

When these rays recombine, they **interfere**. Depending on the film thickness, some wavelengths cancel out (destructive interference) while others reinforce one another (constructive interference). Optical engineering consists in stacking materials with nanometre precision so as to shape these reflection and transmission spectra to order.

Rather than summing by hand the infinite series of multiple reflections inside each layer (Airy's formula), we use a global approach derived from electromagnetism: **the matrix method**. Each layer is modelled by a 2×2 matrix; the effect of a 50-layer stack is nothing more than the product of 50 matrices.



2. Conventions and notation Level 2

So that the document can be read without ambiguity, every parameter, variable and symbol used here is collected below, grouped by physical domain:

2.1 Wave parameters and space-time quantities

SYMBOL	NAME	UNIT	DEFINITION & PHYSICAL MEANING
λ	Wavelength	nm (or $\mu\mathrm{m}$)	Wavelength of the incident light in vacuum (c/f).
ω	Angular frequency	$\mathrm{rad}\cdot\mathrm{s}^{-1}$	$\omega = 2\pi c/\lambda = 2\pi f$, angular frequency of the temporal oscillation.
c	Speed of light	$\mathrm{m}\cdot\mathrm{s}^{-1}$	Speed of light in vacuum ($c = 1/\sqrt{\varepsilon_0\mu_0} \approx 3\times10^8\ \mathrm{m/s}$).
k_0	Vacuum wavenumber	$\mathrm{rad}\cdot\mathrm{m}^{-1}$	$k_0 = \omega/c = 2\pi/\lambda$.
z	Normal spatial coordinate	nm (or m)	Axis pointing **upwards** (↑), with $z = 0$ at the substrate surface.
z_j	Interface height	nm	$z_j = \sum_{k=1}^{j} d_k$, height of the top of layer j ($z_0 = 0$).
θ_0	Angle of incidence	rad or $^\circ$	Angle of the incident beam in air relative to the normal z.
θ_j	Angle of refraction	rad (complex)	Angle inside layer j ($\hat{n}_j \sin\theta_j = n_0 \sin\theta_0$).
S_0	Snell invariant	dimensionless	$S_0 = n_0 \sin\theta_0$, invariant throughout the whole multilayer stack.
k_x	Transverse wave component	$\mathrm{rad}\cdot\mathrm{m}^{-1}$	$k_x = k_0 S_0 = k_0 n_0 \sin\theta_0$ (conserved by translational invariance).
$k_{z,j}$	Normal wave component	$\mathrm{rad}\cdot\mathrm{m}^{-1}$	$k_{z,j} = k_0 \hat{n}_j \cos\theta_j = k_0\sqrt{\hat{n}_j^2 - S_0^2}$.

2.2 Media, materials and layer stack

SYMBOL	NAME	UNIT	DEFINITION & PHYSICAL MEANING
n	Refractive index (real)	dimensionless	Slowing of the phase velocity ($v_\phi = c/n$) inside the material ($n \geq 1$).
κ	Extinction coefficient	dimensionless	Attenuation of the wave by dielectric/ohmic absorption ($\kappa \geq 0$).
$\hat{n}$	Complex refractive index	dimensionless	$\hat{n} = n - i\kappa$ (Macleod convention $e^{+i\omega t}$, which imposes $\mathrm{Im}(\hat{n}) \leq 0$).
n_0, η_0	Index & admittance of air	dimensionless / $\mathcal{Y}_0$	Semi-infinite incident medium at the top ($z \geq z_N$, with $n_0 = 1.0$).
n_{sub}, η_{sub}	Index & admittance of the substrate	dimensionless / $\mathcal{Y}_0$	Semi-infinite exit medium at the base of the stack ($z \leq 0$).
d or d_j	Physical (geometrical) thickness	nm	Metric thickness of the layer ($d_j = z_j - z_{j-1}$).
nd or $n_j d_j$	Optical thickness	nm	Effective optical path travelled by the light ($n \times d$).
δ or δ_j	Optical phase thickness of a layer	rad	$\delta_j = k_{z,j} d_j = \frac{2\pi}{\lambda} \hat{n}_j d_j \cos\theta_j$.
N	Total number of layers	integer	Number of layers in the multilayer (numbered from $j = 1$ at the base to $j = N$ at the top).

2.3 Electromagnetism, fields and matrix formalism (Macleod)

SYMBOL	NAME	UNIT	DEFINITION & PHYSICAL MEANING
$\vec{E}, \vec{H}$	Fields $\vec{E}$ and $\vec{H}$	$\mathrm{V}\cdot\mathrm{m}^{-1}$, $\mathrm{A}\cdot\mathrm{m}^{-1}$	Complex electric and magnetic field vectors.
$E_{\parallel}, H_{\parallel}$	Tangential fields	$\mathrm{V}\cdot\mathrm{m}^{-1}$, $\mathrm{A}\cdot\mathrm{m}^{-1}$	Components parallel to the interface planes (strictly continuous at $z = z_j$).
E_j^+, E_j^-	Upward / downward waves	$\mathrm{V}\cdot\mathrm{m}^{-1}$	Amplitudes of the waves travelling upwards ($+$, reflection) and downwards ($-$).
η_j	Tilted optical admittance	dimensionless (norm.)	Ratio of the tangential fields $H_{\parallel}/E_{\parallel}$ inside layer j.
$\eta_{j,s}$	Admittance in TE (s) polarisation	dimensionless	$\eta_{j,s} = \hat{n}_j \cos\theta_j$ ($\vec{E}$ perpendicular to the plane of incidence).
$\eta_{j,p}$	Admittance in TM (p) polarisation	dimensionless	$\eta_{j,p} = \hat{n}_j / \cos\theta_j$ ($\vec{H}$ perpendicular to the plane of incidence).
M_j	Layer transfer matrix	2×2 matrix	Relates $[E(z_j), H(z_j)]^T = M_j [E(z_{j-1}), H(z_{j-1})]^T$ ($\det M_j = 1$).
M	Overall characteristic matrix	2×2 matrix	Ordered product from bottom to top: $M = M_N M_{N-1} \cdots M_1$.
B, C	Equivalent field vector	dimensionless / admittance	$\begin{pmatrix} B \\ C \end{pmatrix} = M \begin{pmatrix} 1 \\ \eta_{sub} \end{pmatrix}$, total field normalised at the top ($z = z_N$).

2.4 Optical coefficients, energy balances & photometry

SYMBOL	NAME	UNIT	DEFINITION & PHYSICAL MEANING
r, t	Complex Fresnel coefficients	dimensionless (complex)	Amplitudes: $r = \frac{\eta_0 B - C}{\eta_0 B + C}$ (reflection) and $t = \frac{2\eta_0}{\eta_0 B + C}$ (transmission).
R	Energy reflectance	dimensionless ($[0, 1]$)	Fraction of reflected power: $R = \lvert r \rvert^2$.
T	Energy transmittance	dimensionless ($[0, 1]$)	Fraction of transmitted power: $T = \frac{\mathrm{Re}(\eta_{sub})}{\mathrm{Re}(\eta_0)} \lvert t \rvert^2$.
A	Energy absorptance	dimensionless ($[0, 1]$)	Fraction of power absorbed within the layers: $A = 1 - R - T$.
$\vec{S}, S_z$	Poynting vector & normal flux	$\mathrm{W} \cdot \mathrm{m}^{-2}$	Power density crossing the interface: $S_z = \frac{1}{2}\mathrm{Re}(\eta)\lvert E_{\parallel} \rvert^2$.
R_{total}, T_{total}	Coefficients including the back surface	dimensionless	Incoherent photometric coefficients accounting for the second glass surface.

A NOTE ON THE UNITS OF OPTICAL ADMITTANCE (H) AND THE FUNDAMENTAL CONSTANTS

- **Fundamental constants of vacuum:**
  - Vacuum permittivity: $\varepsilon_0 \approx 8.854 \times 10^{-12}\ \mathrm{F} \cdot \mathrm{m}^{-1}$ ($\mathrm{C}^2 \cdot \mathrm{N}^{-1} \cdot \mathrm{m}^{-2}$).
  - Vacuum permeability: $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H} \cdot \mathrm{m}^{-1} \approx 1.257 \times 10^{-6}\ \mathrm{N} \cdot \mathrm{A}^{-2}$.
  - Intrinsic admittance of vacuum: $\mathcal{Y}_0 = \sqrt{\varepsilon_0/\mu_0} \approx \frac{1}{376.73\ \Omega} \approx 2.654 \times 10^{-3}\ \Omega^{-1}$ (siemens, S).
- **Normalised admittance (Macleod convention):** In thin-film theory one works almost systematically with the optical admittance *normalised to that of vacuum*: $\eta = \mathcal{Y}/\mathcal{Y}_0$. Consequently η, $\eta_s = \hat{n}\cos\theta$ and $\eta_p = \hat{n}/\cos\theta$ are handled as **dimensionless quantities** (expressed in "units of refractive index").

*Frame and numbering convention: the z axis points **upwards** ($\uparrow$), in the direction of physical growth during deposition. The plane $z = 0$ is the substrate surface, layer 1 sits directly on the substrate ($0 \le z \le d_1$), and layer N is the last layer deposited, in contact with air at $z = z_N = \sum_{j=1}^{N} d_j$.*

CHOICE OF SPATIAL FRAME: WHY $Z = 0$ AT THE SUBSTRATE, POINTING UPWARDS?

This choice matches exactly the **experimental reality of laboratory deposition** (PVD, ALD, evaporation):

- **Physical growth:** The bare substrate acts as the reference base ($z = 0$). Thickness accumulates upwards as the layers are deposited ($j = 1, 2, \ldots, N$).
- **Consistency with Macleod's matrix formalism:** The boundary conditions are initialised on the substrate at the base ($z = 0$): $\begin{pmatrix} E(0) \\ H(0) \end{pmatrix} = \begin{pmatrix} 1 \\ \eta_{sub} \end{pmatrix}$. Working upwards layer by layer, one multiplies successively by M_1 , M_2, ..., up to M_N. At the top of the stack ($z = z_N$), the total field in contact with the incident air is given directly by the characteristic vector $\begin{pmatrix} B \\ C \end{pmatrix} = M_N \cdots M_1 \begin{pmatrix} 1 \\ \eta_{sub} \end{pmatrix}$.
- **Direction of illumination:** The incident light arrives from the air ($z \geq z_N$) and propagates downwards (towards decreasing z, i.e. towards the substrate).

FUNDAMENTAL PROOF: WHY IS $\mathrm{IM}(\hat{N}) \leq 0$ WITH THE $E^{-I\Omega T}$ CONVENTION?

In harmonic electromagnetism, the choice of time dependence fixes unambiguously the sign of the imaginary part of the refractive index. Proof:

1. Under the **Macleod / engineering** convention, the fields vary as $e^{+i\omega t}$. A plane wave propagating in the forward direction ($+z$) is written:

$$\vec{E}(z,t) = \vec{E}_0\, e^{i(\omega t - kz)} \quad \text{with} \quad k = k_0 \hat{n} = \frac{\omega}{c}\hat{n}$$

2. Let us write the complex index with its real part n and its imaginary part $\mathrm{Im}(\hat{n})$:

$$\vec{E}(z,t) = \vec{E}_0\, e^{i(\omega t - k_0(n + i\,\mathrm{Im}(\hat{n}))z)} = \vec{E}_0 \underbrace{e^{k_0 \mathrm{Im}(\hat{n})z}}_{\text{Spatial attenuation}} \underbrace{e^{i(\omega t - k_0 n z)}}_{\text{Oscillating phase}}$$

3. For a passive **absorbing** medium, the amplitude must decay exponentially as $e^{-k_0 \kappa z}$ (Beer-Lambert law, with extinction coefficient $\kappa \geq 0$).

By direct identification of the amplitude factor:

$$\mathrm{Im}(\hat{n}) = -\kappa \leq 0 \implies \mathbf{\hat{n} = n - i\kappa} \quad \mathbf{(\kappa \geq 0)}$$

Warning: Were you to use a plus sign by mistake ($\hat{n} = n + i\kappa$) together with $e^{+i\omega t}$, the spatial factor would become $e^{+k_0 \kappa z}$, simulating **unphysical exponential amplification** (spontaneous creation of infinite energy inside a passive material).

SCIENTIFIC BACKGROUND: THE OTHER CONVENTION ($E^{-I\Omega T}$) IN THEORETICAL PHYSICS

It is essential to know that a second convention coexists in the scientific literature (quantum physics, fundamental optics, the treatises of Born & Wolf, Jackson, Landau):

In this second convention, time is taken as $e^{-i\omega t}$. The wave propagating towards $+z$ is then written $\vec{E}(z,t) = \vec{E}_0 e^{-i(\omega t - kz)} = \vec{E}_0 e^{i(kz - \omega t)}$. To ensure attenuation as $e^{-k_0 \kappa z}$, the complex index is then written with a **plus sign**:

$$\mathbf{\hat{n} = n + i\kappa} \quad \mathbf{(Theoretical\text{-}physics\ convention\ e^{-i\omega t})}$$

CRITERION	THIN-FILM CONVENTION (MACLEOD)	THEORETICAL-PHYSICS CONVENTION (BORN & WOLF)
Time dependence	$e^{+i\omega t}$ (or $\partial_t \to +i\omega$)	$e^{-i\omega t}$ (or $\partial_t \to -i\omega$)
Travelling wave ($+z$)	$\exp[i(\omega t - kz)]$	$\exp[-i(\omega t - kz)] = \exp[i(kz - \omega t)]$
Complex index $\hat{n}$	$\hat{\mathbf{n}} = \mathbf{n} - \mathbf{i}\kappa$	$\hat{\mathbf{n}} = \mathbf{n} + \mathbf{i}\kappa$
Imaginary part $\mathrm{Im}(\hat{n})$	**Strictly negative** ($-\kappa \leq 0$)	**Strictly positive** ($+\kappa \geq 0$)
Dominant usage	Thin-film software (OptiLayer, TFCalc, the Python code of this course)	Classical electrodynamics, quantum mechanics, nonlinear optics

Advice for the engineer: Databases (e.g. `refractiveindex.info`) always provide k (or κ) as a positive real number. Within our Macleod formalism, you must **always feed in** `n_complex = n - 1j * k`.

3. From Maxwell's equations to the transfer matrix Level 2

This section establishes rigorously, from the first principles of electromagnetism, how Maxwell's equations reduce to the Helmholtz equation, and then how the boundary conditions together with the outgoing-wave condition allow the 2×2 transfer matrix of a thin film to be formulated.

3.1 Maxwell's equations in the harmonic regime

We adopt **Macleod's** time convention, in which every field oscillates at the angular frequency ω as $e^{+i\omega t}$. For a real electric field $\vec{\mathcal{E}}_{phys}(\vec{r}, t) = \mathrm{Re}[\vec{E}(\vec{r})\, e^{i\omega t}]$, time differentiation becomes a simple algebraic product: $\frac{\partial}{\partial t} \longrightarrow +i\omega$.

In a linear, homogeneous, isotropic (LHI) and non-magnetic ($\mu_r = 1$) dielectric or conducting medium, free of macroscopic free charges ($\rho = 0$), Maxwell's equations take the complex form:

$$\nabla \cdot \vec{D} = 0 \quad \text{with} \quad \vec{D} = \varepsilon_0 \varepsilon_r \vec{E} \qquad \text{(Maxwell-Gauss)}$$

$$\nabla \cdot \vec{B} = 0 \quad \text{with} \quad \vec{B} = \mu_0 \vec{H} \qquad \text{(Maxwell-Thomson)}$$

$$\nabla \times \vec{E} = -i\omega\mu_0 \vec{H} \qquad \text{(Maxwell-Faraday)}$$

$$\nabla \times \vec{H} = \vec{j} + i\omega\vec{D} = (\sigma + i\omega\varepsilon_0\varepsilon_r)\vec{E} = i\omega\varepsilon_0\hat{\varepsilon}_r\vec{E} \qquad \text{(Maxwell-Ampère)}$$

The relative complex permittivity $\hat{\varepsilon}_r$ unifies the dielectric constant ε_r and the ohmic conductivity σ. It is directly related to the **complex refractive index** $\hat{n} = n - i\kappa$:

$$\hat{\varepsilon}_r = \varepsilon_r - i\frac{\sigma}{\omega\varepsilon_0} = \hat{n}^2 = (n - i\kappa)^2 = (n^2 - \kappa^2) - 2in\kappa$$

3.2 Wave equation and Helmholtz equation

Applying the curl operator to the Maxwell-Faraday equation and using the vector identity $\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \Delta\vec{E}$ with $\nabla \cdot \vec{E} = 0$ (homogeneous medium), one obtains:

$$\nabla(\underbrace{\nabla \cdot \vec{E}}_{=0}) - \Delta\vec{E} = -i\omega\mu_0(\nabla \times \vec{H}) = -i\omega\mu_0(i\omega\varepsilon_0\hat{n}^2\vec{E}) = \omega^2\mu_0\varepsilon_0\hat{n}^2\vec{E}$$

Since the speed of light in vacuum is $c = 1/\sqrt{\varepsilon_0\mu_0}$ and the vacuum wavenumber is $k_0 = \omega/c = 2\pi/\lambda$, we arrive at the **vector Helmholtz equation**:

$$\Delta\vec{E}(\vec{r}) + k_0^2\hat{n}^2\vec{E}(\vec{r}) = 0$$

Consider an incident plane wave propagating in the plane of incidence (x, z), where z is the axis perpendicular to the layers and x the axis parallel to the interfaces. By translational invariance along the transverse axis y, we have $\partial/\partial y = 0$. The spatial dependence along x is shared by every layer and is set by the incident medium:

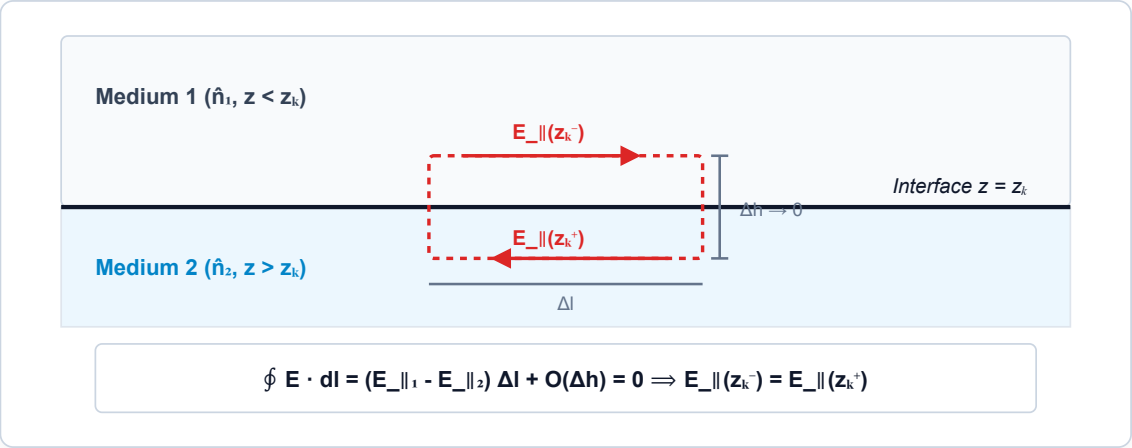
$$\vec{E}(x, z) = \vec{\mathcal{E}}(z)\, e^{-ik_x x} \quad \text{with} \quad k_x = k_0 n_0 \sin\theta_0 = k_0 S_0 \quad \text{(Snell invariant)}$$

The Helmholtz equation then reduces to a one-dimensional scalar differential equation in the coordinate z for each component:

$$\frac{d^2\vec{\mathcal{E}}(z)}{dz^2} + k_z^2\vec{\mathcal{E}}(z) = 0 \quad \text{with} \quad k_z = \sqrt{k_0^2\hat{n}^2 - k_x^2} = k_0\sqrt{\hat{n}^2 - S_0^2} = k_0\hat{n}\cos\theta$$

3.3 Continuity conditions at interfaces (boundary conditions)

At each interface $z = z_k$ separating two media of indices $\hat{n}_1$ and $\hat{n}_2$, Maxwell's equations impose strict continuity relations. Integrating Maxwell-Faraday and Maxwell-Ampère over an infinitesimal rectangular contour straddling the interface (width Δl, height $\Delta h \to 0$), the magnetic flux and the current threading the loop both tend to zero:



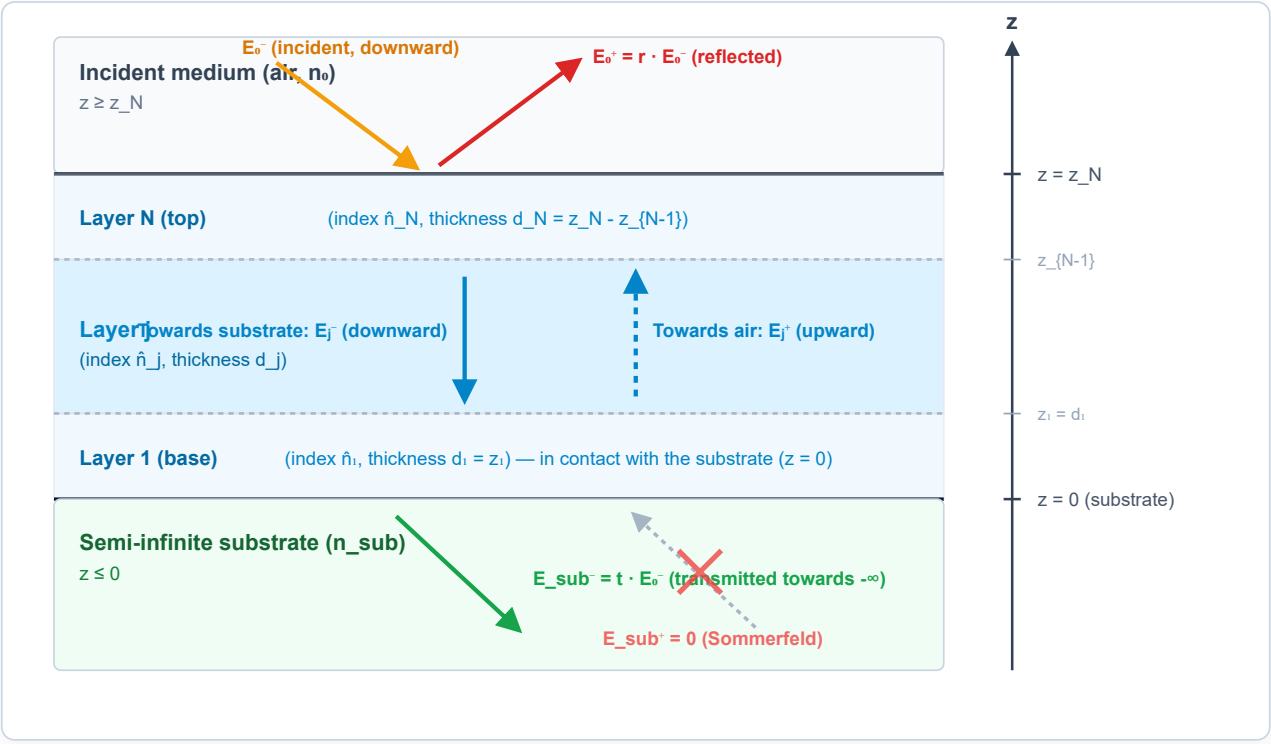
In the absence of free surface charges and currents, the **tangential components** of the electric and magnetic fields are strictly continuous:

$$E_{\parallel}(z_k^-) = E_{\parallel}(z_k^+) \qquad \text{and} \qquad H_{\parallel}(z_k^-) = H_{\parallel}(z_k^+)$$

3.4 Travelling waves and the outgoing-wave condition (Sommerfeld)

Within each homogeneous layer j (bounded by its base z_{j-1} and its top $z_j = z_{j-1} + d_j$), the general solution of the 1D Helmholtz equation is the exact superposition of a wave propagating downwards (towards the substrate, in the direction of decreasing z, written E_j^-) and a reflected wave propagating upwards (towards the air, in the direction of increasing z, written E_j^+):

$$E_{\parallel,j}(z) = E_j^+ \, e^{-ik_{z,j}(z-z_{j-1})} + E_j^- \, e^{+ik_{z,j}(z-z_j)}$$



THE SOMMERFELD RADIATION CONDITION (OUTGOING WAVE)

In the semi-infinite substrate ($z \leq 0$), the transmitted wave travels away downwards ($z \to -\infty$). In the absence of any light source located at $-\infty$, **no backward wave can come back up from the depths of the substrate**:

$$\mathbf{E}_{\mathbf{sub}}^{+} = \mathbf{0}$$

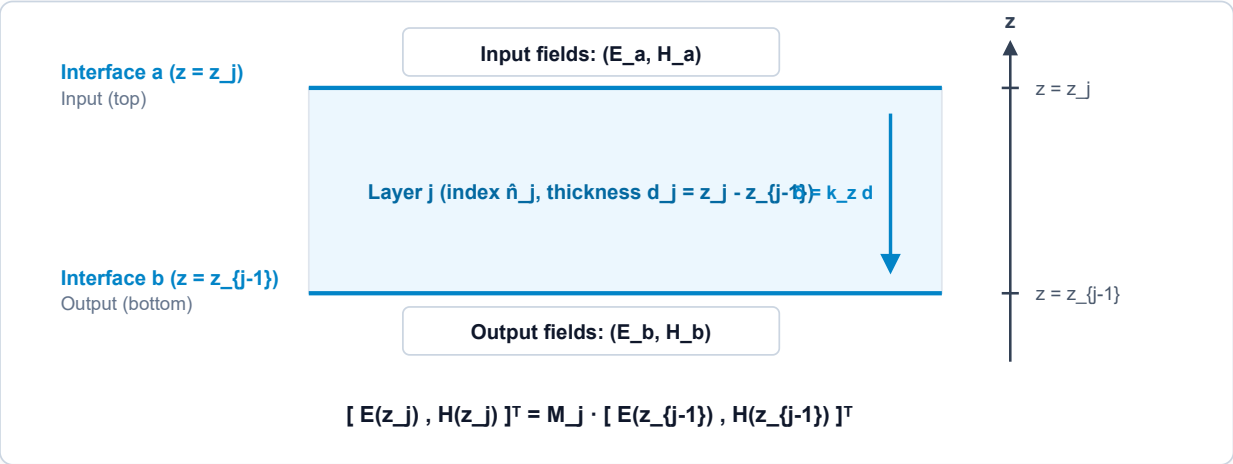
It is this essential boundary condition at infinity that closes the system mathematically and makes it possible to compute the Fresnel coefficients r and t uniquely.

3.5 Analytical derivation of the 2×2 characteristic matrix

To relate the tangential magnetic field $H_{\parallel}$ to the tangential electric field $E_{\parallel}$, we use the Maxwell-Faraday equation $\nabla \times \vec{E} = -i\omega\mu_0\vec{H}$:

- **TE (s) polarisation:** $\vec{E} = (0, E_y, 0)$. Then $H_x = -\frac{1}{i\omega\mu_0}\frac{\partial E_y}{\partial z} = \eta_s(E^- e^{+ik_z z} - E^+ e^{-ik_z z})$ with the admittance $\eta_s = \hat{n}\cos\theta$.
- **TM (p) polarisation:** $\vec{H} = (0, H_y, 0)$. Then $E_x = \frac{1}{i\omega\varepsilon_0\hat{n}^2}\frac{\partial H_y}{\partial z}$ with the admittance $\eta_p = \frac{\hat{n}}{\cos\theta}$.

Consider a layer j of thickness d lying between its **upper input interface** a ($z = z_j$) and its **lower output interface** b ($z = z_{j-1} = z_j - d$):



1. **At the upper input interface ($z = z_j$):**

$$E_a = E^- + E^+ \qquad \text{and} \qquad H_a = \eta(E^- - E^+)$$

which allows the wave amplitudes to be expressed in terms of the total input fields:

$$E^- = \frac{1}{2}\left(E_a + \frac{H_a}{\eta}\right) \qquad \text{and} \qquad E^+ = \frac{1}{2}\left(E_a - \frac{H_a}{\eta}\right)$$

2. **At the lower output interface ($z = z_{j-1} = z_j - d$), writing the phase thickness as $\delta = k_z d$:**

$$E_b = E^- e^{-i\delta} + E^+ e^{+i\delta} = \frac{1}{2}\left(E_a + \frac{H_a}{\eta}\right)e^{-i\delta} + \frac{1}{2}\left(E_a - \frac{H_a}{\eta}\right)e^{+i\delta} = E_a \cos\delta - \frac{i}{\eta} H_a \sin\delta$$

$$H_b = \eta\left(E^- e^{-i\delta} - E^+ e^{+i\delta}\right) = -i\eta E_a \sin\delta + H_a \cos\delta$$

3. **Matrix inversion, to express the input fields (E_a, H_a) in terms of the output fields (E_b, H_b):**

$$\begin{pmatrix} E(z_j) \\ H(z_j) \end{pmatrix} = \begin{pmatrix} \cos\delta & \frac{i}{\eta}\sin\delta \\ i\eta\sin\delta & \cos\delta \end{pmatrix} \begin{pmatrix} E(z_{j-1}) \\ H(z_{j-1}) \end{pmatrix} = M_j \begin{pmatrix} E(z_{j-1}) \\ H(z_{j-1}) \end{pmatrix}$$

FUNDAMENTAL PROPERTY: UNIMODULARITY

The determinant of the characteristic matrix of a layer is rigorously equal to unity:

$$\det(M) = (\cos\delta)(\cos\delta) - \left(\frac{i}{\eta}\sin\delta\right)(i\eta\sin\delta) = \cos^2\delta + \sin^2\delta = 1$$

This **unimodularity** property ($\det M = 1$) guarantees electromagnetic reciprocity and the strict conservation of energy flux in transparent media.

SECTION 3 IN A NUTSHELL — 4 KEY POINTS

- **Harmonic convention:** time dependence $e^{+i\omega t} \implies$ complex index $\hat{n} = n - i\kappa$ with $\mathrm{Im}(\hat{n}) \leq 0$.
- **Continuity at interfaces:** the tangential components $E_{\parallel}$ and $H_{\parallel}$ are strictly continuous at every boundary.
- **Elementary matrix:** $M_j = \begin{pmatrix} \cos\delta_j & \frac{i}{\eta_j}\sin\delta_j \\ i\eta_j \sin\delta_j & \cos\delta_j \end{pmatrix}$ relates the input at the top (z_j) to the output at the base (z_{j-1}), with $\det(M_j) = 1$.
- **Sommerfeld condition:** in the semi-infinite substrate ($z \leq 0$), no wave comes back up from $-\infty$ ($E^+_{sub} = 0$).

4. Transfer-matrix method (TMM) Level 2

4.1 Characteristic matrix of a layer j

For a layer of complex index $\hat{n}_j$ and thickness d_j, the matrix M_j reads:

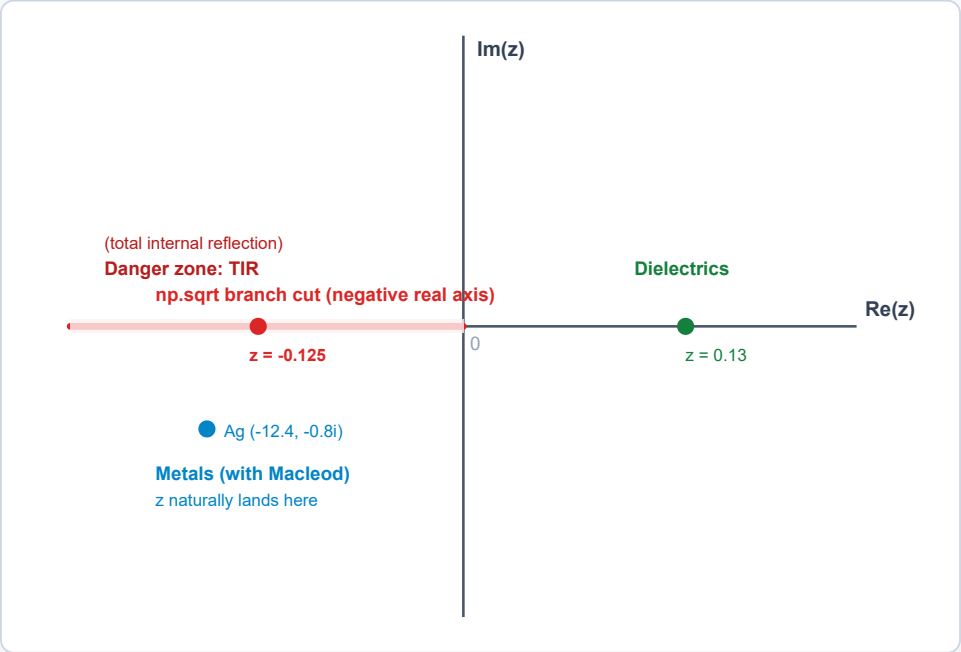
$$M_j = \begin{pmatrix} \cos\delta_j & \frac{i}{\eta_j}\sin\delta_j \\ i\eta_j \sin\delta_j & \cos\delta_j \end{pmatrix}$$

where δ_j is the accumulated **optical phase**: $\delta_j = \frac{2\pi}{\lambda}(\hat{n}_j) d_j \cos\theta_j$.

GOING FURTHER — **The branch-cut pitfall for evanescent waves**

Context: Snell's law gives $S_0 = n_0 \sin\theta_0$. To obtain the propagation angle inside layer j, one uses the trigonometric identity $\cos\theta_j = \frac{\sqrt{\hat{n}_j^2 - S_0^2}}{\hat{n}_j}$. The pitfall lies in the numerical evaluation of the square root $\sqrt{z}$.

The function $f(z) = \sqrt{z}$ is multivalued over $\mathbb{C}$. In Python/NumPy, `np.sqrt(z)` returns by default the root with a real part ≥ 0 and places a branch cut along the negative real axis.



Physical attenuation condition

The electric field inside layer j evolves as $E_j(z,t) \propto \exp[-ik_{z,j}z]$ with $k_{z,j} = \frac{2\pi}{\lambda}\hat{n}_j \cos\theta_j$. For the amplitude to **decay** as it penetrates the material (absorption or evanescent wave), one must imperatively satisfy:

$$\mathrm{Im}(\hat{n}_j \cos\theta_j) \leq 0 \quad \text{(with the } n - i\kappa \text{ convention)}$$

The critical case: total internal reflection (TIR)

Contrary to a widespread belief, metals cause no trouble with the Macleod convention: $\hat{n}_{Ag} = 0.12 - 3.45i$ gives $z \approx -12.4 - 0.8i$ (lower half-plane). `np.sqrt(z)` returns $0.11 - 3.52i$, which does satisfy $\mathrm{Im} \leq 0$.

The real problem arises during **total internal reflection** in a dielectric (e.g. light passing from glass $n = 1.5$ into air $n = 1.0$ at 45°). One then has $S_0 = 1.06$ and $z = n^2 - S_0^2 = -0.125$ (purely real and negative). In this specific case, `np.sqrt(-0.125 + 0j)` returns `+0.353j`. **The imaginary part is positive!** The simulated wave will grow exponentially instead of decaying.

It is therefore crucial to force the correct sign manually in the code for evanescent waves (see Section 10):

```
if n_cos_th.imag > 0:
    n_cos_th = -n_cos_th  # Force physical attenuation
```

4.2 Overall matrix of the stack

The overall system matrix M is the matrix product of the matrices of each layer. The multiplication runs from the base on the substrate ($z = 0$, layer 1) up to the top in contact with the air ($z = z_N$, layer N):

$$\begin{pmatrix} E(z_N) \\ H(z_N) \end{pmatrix} = M_N \cdot M_{N-1} \cdots M_1 \begin{pmatrix} E(0) \\ H(0) \end{pmatrix} = M \begin{pmatrix} E(0) \\ H(0) \end{pmatrix}$$

$$\text{with} \quad M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} = M_N \cdot M_{N-1} \cdots M_1$$

5. Polarisation and optical admittance Level 2

The optical admittance η_j characterises the ratio of the tangential fields ($H_\parallel / E_\parallel$). At oblique incidence it depends on the polarisation of the light:

POLARISATION	PHYSICAL DESCRIPTION	ADMITTANCE H_J
TE (s)	Transverse electric field (parallel to the interface)	$\eta_{j,s} = \hat{n}_j \cos\theta_j$
TM (p)	Transverse magnetic field (parallel to the interface)	$\eta_{j,p} = \dfrac{\hat{n}_j}{\cos\theta_j}$

6. Fresnel coefficients and powers (R, T) Level 2

For the complete stack, the characteristic vector $\begin{pmatrix} B \\ C \end{pmatrix}$ is defined by multiplying the overall matrix M by the vector of substrate boundary conditions:

$$\begin{pmatrix} B \\ C \end{pmatrix} = M \cdot \begin{pmatrix} 1 \\ \eta_{sub} \end{pmatrix}$$

The amplitude coefficients in reflection (r) and transmission (t) read:

$$r = \frac{\eta_0 B - C}{\eta_0 B + C} \qquad \text{and} \qquad t = \frac{2\eta_0}{\eta_0 B + C}$$

From these follow the experimentally measurable quantities, namely the reflectance R and the transmittance T (with energy conservation $R + T + A = 1$, where A is the absorption):

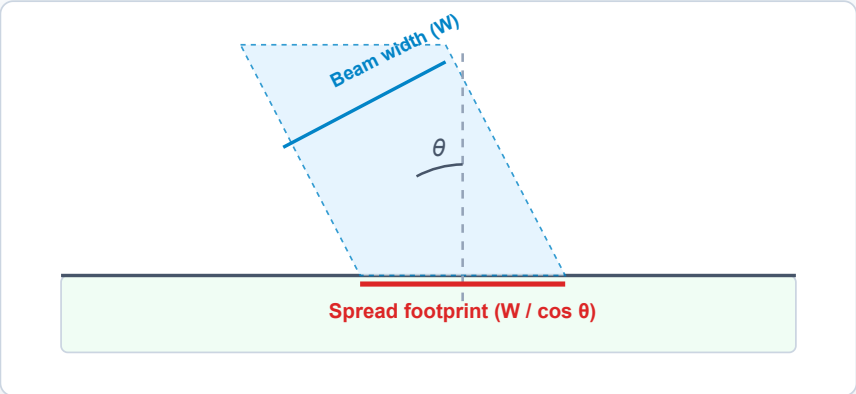
$$R = |r|^2 \qquad \text{and} \qquad T = \frac{\mathrm{Re}(\eta_{sub})}{\mathrm{Re}(\eta_0)} |t|^2$$

🎓 GOING FURTHER — **Why is $T \neq |t|^2$? The Poynting flux at oblique incidence**

A classic student mistake is to write $T = |t|^2$. In fact the transmittance is the ratio of **power densities per unit area** (tied to the flux of the Poynting vector $\vec{S}$), not merely the ratio of the squared electric fields.

The power carried by a wave is $|\vec{S}| = \frac{1}{2}\mathrm{Re}(\hat{n})|E|^2$. But energy conservation at the interface depends on the energy that *actually crosses* that surface, that is, the **normal** component of the flux: S_z.

At oblique incidence (angle θ), a beam of spatial width W "spreads out" over the interface to occupy a footprint of size $W / \cos\theta$. The power per unit area of the interface therefore decreases geometrically by a factor $\cos\theta$.



Mathematically, projecting this flux onto the normal gives:

$$S_z = |\vec{S}| \cos\theta = \frac{1}{2}\mathrm{Re}\Big(\underbrace{\hat{n}\cos\theta}_{\text{Admittance } \eta_s} \Big) |E_{\parallel}|^2$$

Note the term $\hat{n}\cos\theta$: it corresponds *exactly* to the definition of the optical admittance η_s in TE (s) polarisation. The admittance η is therefore not merely a matrix-algebra convenience; it intrinsically embeds the material index and the geometrical projection of the beam (the same proof holds for TM with η_p).

The reflectance remains simply $R = |r|^2$ because the incident and reflected waves live in the same medium (the admittance cancels out in the ratio). But the transmitted wave enters a new substrate, hence the transmission correction factor involving the real parts of the admittances.

SECTION 6 IN A NUTSHELL — 3 ESSENTIAL FORMULAS

- **Characteristic vector:** $\begin{pmatrix} B \\ C \end{pmatrix} = M_{sys} \begin{pmatrix} 1 \\ \eta_{sub} \end{pmatrix} = M_N \cdot M_{N-1} \cdots M_1 \begin{pmatrix} 1 \\ \eta_{sub} \end{pmatrix}$
- **Reflectance & transmittance:** $R = |r|^2 = \left| \frac{\eta_0 B - C}{\eta_0 B + C} \right|^2$ and $T = \frac{\mathrm{Re}(\eta_{sub})}{\mathrm{Re}(\eta_0)} |t|^2 = \frac{\mathrm{Re}(\eta_{sub})}{\mathrm{Re}(\eta_0)} \left| \frac{2\eta_0}{\eta_0 B + C} \right|^2$
- **Conservation balance:** $R + T + A = 1$ (where $A = 1 - R - T$ quantifies the net thermal absorption within the layers).

7. Substrate of finite thickness Level 2

The preceding equations assume a semi-infinite exit substrate. In reality a glass plate has a back surface. Because of the large substrate thickness (millimetric, $\gg \lambda$), spatial optical coherence is lost. We therefore no longer sum complex amplitudes (interference), but the **intensities** of the multiple incoherent bounces inside the glass.

General formula for a plate with an incoherent back surface

1. First compute the coated surface illuminated from the air: R_{front} and T_{front}.
2. Then compute the same surface seen from inside the substrate: R'_{front} and T'_{front}.
3. Finally take the bare back surface: R_{back} and T_{back}.

We then obtain:

$$R_{total} = R_{front} + \frac{T_{front} \cdot T'_{front} \cdot R_{back}}{1 - R'_{front} \cdot R_{back}}$$

$$T_{total} = \frac{T_{front} \cdot T_{back}}{1 - R'_{front} \cdot R_{back}}$$

Worth remembering: in passive linear optics, power transmittance is reciprocal. In other words, provided it is properly defined and properly normalised, $T'_{front} = T_{front}$. What can change with the direction of illumination is mainly R and A.

Important: for a finite substrate, T_{total} is not the same thing as T_{front}. The complete formula above must be applied; otherwise a physics error is committed.

Quick example: bare glass reflects about 4% per surface ($R_{front} = R'_{front} = R_{back} \approx 0.04$). The total reflectance of a single pane is then about 7.7%. The total transmittance must likewise be recomputed with the finite-substrate formula.

CONCEPTUAL CLARIFICATION: TRANSMITTED POWER AND ABSORPTION

The presence of an absorbing medium is not by itself enough to create a difference between T and T' when the **power transmittance** is properly compared. The real subtlety lies elsewhere: absorption modifies the balance $R + T + A = 1$ and can change R and A depending on the direction of illumination, but not the correctly defined power transmission of a reciprocal system.

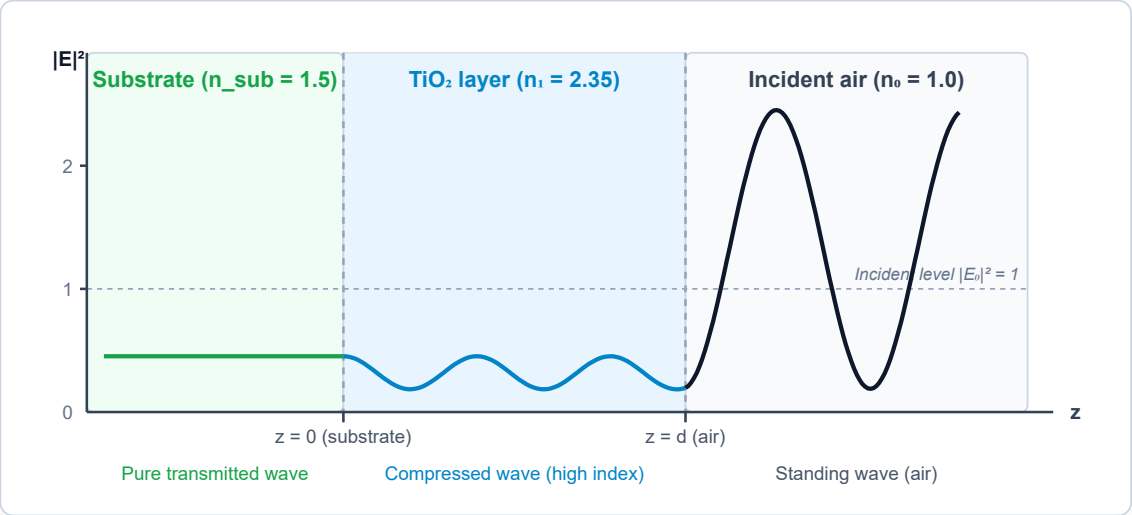
8. [Advanced] Electric-field intensity profile $|E(z)|^2$ Level 3

Computing the intensity profile inside the stack is fundamental, in particular for assessing laser-induced damage thresholds (LIDT) or optimising absorption in solar cells.

Method: The field transmitted into the substrate at $z = 0$ is known (it equals the transmitted amplitude t). One can then work mathematically upwards along the z axis (in the direction of layer growth, from $z = 0$ to the top $z = z_N$) by multiplying by the local partial matrices evaluated at height z:

$$\begin{pmatrix} E(z) \\ H(z) \end{pmatrix} = \begin{pmatrix} \cos\delta(z) & \frac{i}{\eta_j}\sin\delta(z) \\ i\eta_j \sin\delta(z) & \cos\delta(z) \end{pmatrix} \begin{pmatrix} E(0) \\ H(0) \end{pmatrix}$$

where $\delta(z) = \frac{2\pi}{\lambda}\hat{n}_j z\cos\theta_j$ is the phase accumulated since the substrate interface.



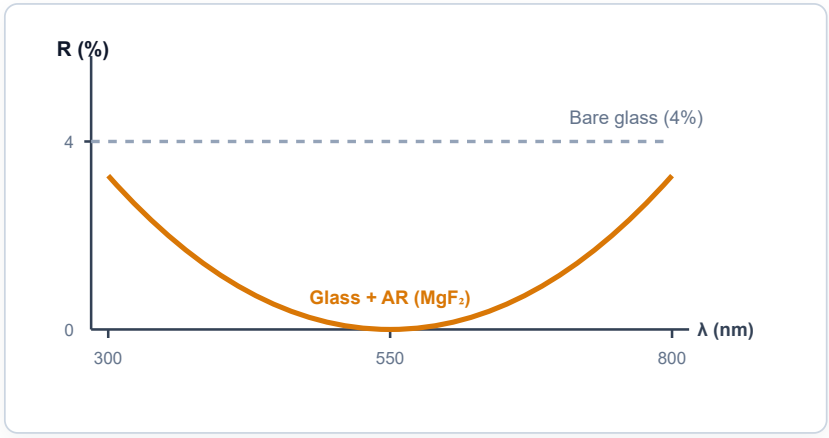
Intensity profile $|E(z)|^2$ computed by TMM for a TiO2 layer ($n = 2.35$) on a substrate ($n_{sub} = 1.5$), illuminated from the air on the right ($z \geq d$).

Physical note: the z axis starts at the substrate surface ($z = 0$) and increases towards the surface in contact with the air ($z = d$).

9. Worked numerical examples Levels 1 & 2

Case 1: quarter-wave anti-reflection (AR) coating ($\lambda/4$)

The goal is to cancel the $\sim 4\%$ of stray reflection from the glass. A thickness is deposited such that the round-trip phase shift is π (fully destructive interference). The required optical thickness is $nd = \lambda/4$.



The reflection vanishes completely if the film index satisfies $n_{film} = \sqrt{n_{air} \cdot n_{glass}}$. For standard glass one would want $n \approx 1.22$. Magnesium fluoride (MgF2, $n = 1.38$) is the closest material commonly used in classical optics to approach this ideal.

ENGINEERING ASIDE — A NOMINAL OPTIMUM IS NOT YET A DESIGN

A thickness computed to six decimal places is not an industrial prescription. Manufacturing introduces finite resolution, random layer-by-layer errors and correlated errors, for instance a calibration bias of the quartz monitor. The decision must therefore follow the complete chain set out below.

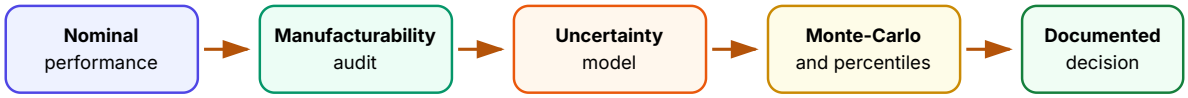
Experimental thread — imposed Fresnel prescription.

`BK7 / H L 2H L H L H L 2H L H L H L 2H L H L / Air`

The stack comprises exactly **18 layers**, numbered from the BK7 substrate towards the air. In pre-manufacturing, BFGS may modify only layers **16, 17 and 18**; the first fifteen define the architecture that will be deposited. Performance is assessed over the useful window, not over the whole spectrum:

$$Q_r = \frac{2\,\overline{T}_{535-565}}{\overline{T}_{460-500} + \overline{T}_{620-680}}, \qquad \overline{T}_{pass} = \overline{T}_{535-565}.$$

These averages must first be measured separately on T and then combined by the student. The optimisation does not allow the number of layers to be changed, nor the result to be "improved" by releasing the first fifteen. The qualification, on the other hand, perturbs all **18 layers actually deposited**: a layer frozen at the design stage is not a layer free of manufacturing error.



Nominal manufacturability. One checks at minimum the number of layers and materials, the deposition order, d_{min}, d_{max}, the total thickness and the compatibility of every value with the resolution of the process. A layer below the adopted continuity threshold makes the prescription unmanufacturable, however excellent its nominal spectrum may be.

Single teaching model. Each deposited layer receives an independent Gaussian error of standard deviation $\sigma = \Delta d/2$. The pseudo-random seed is fixed so as to make studies reproducible and A/B comparisons paired. The layer lock belongs solely to the BFGS optimisation: a layer frozen during design is still physically deposited and is therefore perturbed during qualification.

INTERPRETING ΔD

Δd is the severity parameter of the model, not an absolute bound: with $\sigma = \Delta d/2$, about 95% of the idealised errors fall within $\pm\Delta d$. This model represents neither a common calibration bias, nor the quantisation of the quartz monitor, nor a correlated drift of the deposition rate.

Yield and uncertainty. If x realisations out of N satisfy both manufacturability and the optical specification, the estimated yield is $\hat{p} = x/N$. The 95% Wilson interval is:

$$CI_{95\%} = \frac{\hat{p} + z^2/(2N) \;\pm\; z\sqrt{\hat{p}(1-\hat{p})/N + z^2/(4N^2)}}{1 + z^2/N}, \qquad z = 1.96.$$

The yield must be reported *together with* this interval. A displayed value of 100% over a finite number of draws never means "zero risk".

INDICATOR	ENGINEERING QUESTION	CORRECT USE
Mean ± standard deviation	Where does the population sit and how dispersed is it?	Description, not a decision on its own
P05	What low performance is still exceeded by 95% of realisations?	Quantity to maximise: Q_r, T_{pass}
P95	What high cost is exceeded by only 5% of realisations?	Quantity to minimise: $\overline{R}$, RMSE
Wilson 95% CI	What is the sampling uncertainty on the yield?	Compare the lower bound with the requirement

LIMIT OF VALIDITY

A Monte-Carlo study is only as good as its assumptions. It covers neither an index error, nor roughness, nor lateral inhomogeneity, nor temperature, nor ageing, unless these phenomena are explicitly introduced. A teaching qualification is therefore a **decision conditional on the model**, not a process certification.

Case 2: five-layer validation filter (with absorbing metal)

This demanding test case validates the Python code, since it includes oblique angles, polarisation and absorption. Above all it serves to check two things:

- energy conservation $R + T + A = 1$,
- the reciprocity of the **power transmittance**: $T = T'$, even in an absorbing medium, provided the comparison is made with the same physical normalisation.

- **Incident medium:** air ($n = 1.0$) | **Substrate:** glass ($n = 1.52$) at $\lambda = 550$ nm.
- **Stack (numbered from the substrate towards the air):**

1. TiO_2 ($n = 2.35$), 85 nm
2. SiO_2 ($n = 1.46$), 110 nm
3. **Ag ($n = 0.12 - 3.45i$), 15 nm** (absorbing metal layer)
4. TiO_2 ($n = 2.35$), 85 nm
5. SiO_2 ($n = 1.46$), 90 nm

VALIDATED REFERENCE TABLE

The table below collects the exact theoretical results for the coated surface alone (semi-infinite substrate) and for the complete plate (accounting for the multiple incoherent bounces on the bare glass back surface):

ANGLE AND POLARISATION	FRONT SURFACE ALONE (SEMI-INFINITE)			PLATE WITH BARE BACK SURFACE	
	REFLECTANCE R	TRANSMITTANCE T	ABSORPTION A	R_{TOTAL}	T_{TOTAL}
0° (normal)	0.357319	0.556823	0.085858	0.370750	0.542352
45° (s / TE polarisation)	0.586690	0.324859	0.088451	0.597575	0.312870
45° (p / TM polarisation)	0.482173	0.446639	0.071187	0.484049	0.444613

10. Numerical implementation (Python) Level 2

The code below is the mathematical summary of the theory presented here. It accounts for absorption, angle of incidence and polarisations, and corrects the complex square-root problem for evanescent waves (see § 4.1).

🎓 GOING FURTHER — **Show the didactic Python TMM script**

```
import numpy as np

def compute_tmm(n_inc, n_exit, n_layers, d_layers, wl, theta0=0.0, pol='s'):
    """
    Compute the optical properties of a thin-film stack (TMM).
```

```
    Arguments:
      n_inc, n_exit : Complex indices of the incident medium (air) and the
substrate.
      n_layers      : List of the complex indices of the layers (n - i*k).
      d_layers      : List of the physical thicknesses of the layers (in nm).
      wl            : Working wavelength (in nm).
      theta0        : Angle of incidence in the incident medium (in radians).
      pol           : Polarisation ('s' for TE, 'p' for TM).

    Returns:
      R (reflectance), T (transmittance), A (absorption)
    """

    # ========================================================================
    # Step 1: Enforcing the Macleod convention (guarantees n - ik)
    # ========================================================================
    # In thin-film optics the wave is formally written exp(i(wt - kz)).
    # For the medium to absorb energy (rather than amplify it), the imaginary
    # part of the complex index must be strictly negative.
    def enforce_macleod(n):
        return n if n.imag <= 0 else n.real - 1j * n.imag

    n_inc = enforce_macleod(n_inc)
    n_exit = enforce_macleod(n_exit)
    n_layers = [enforce_macleod(n) for n in n_layers]

    # Vacuum wavenumber
    k0 = 2 * np.pi / wl

    # ========================================================================
    # Step 2: Snell's law and complex angles (TIR correction)
    # ========================================================================
    # The Snell invariant (n * sin(theta)) is conserved everywhere.
    S0 = n_inc * np.sin(theta0)

    def safe_cos_th(n_val):
        # We use the classic trigonometric identity:
        # cos(theta) = sqrt(n^2 - (n*sin(theta))^2) / n
        n_cos_th = np.sqrt(n_val**2 - S0**2 + 0j)

        # BEWARE OF THE NUMERICAL PITFALL (complex branch cut):
        # In case of total internal reflection (TIR) or an evanescent wave,
        # numpy may return a positive imaginary part, which simulates an
        # unphysical energy gain. We force the physical decay:
        if n_cos_th.imag > 0:
            n_cos_th = -n_cos_th
        return n_cos_th / n_val

    # Cosines (complex under TIR/absorption) for each medium
    cos_inc = np.cos(theta0)
    cos_exit = safe_cos_th(n_exit)
    cos_layers = [safe_cos_th(n) for n in n_layers]
```

```
    # ========================================================================
    # Step 3: Computing the optical admittances (eta)
    # ========================================================================
    # The admittance relates the magnetic field to the electric field (H = eta
* E)
    def get_eta(n_val, cos_th, pol):
        if pol.lower() == 's':
            # TE polarisation (transverse E field): projection by
multiplication
            return n_val * cos_th
        else:
            # TM polarisation (transverse H field): projection by division
            return n_val / cos_th

    eta_inc = get_eta(n_inc, cos_inc, pol)
    eta_exit = get_eta(n_exit, cos_exit, pol)
    eta_layers = [get_eta(n_layers[i], cos_layers[i], pol) for i in
range(len(n_layers))]

    # ========================================================================
    # Step 4: Matrix multiplication (from the substrate towards the air)
    # ========================================================================
    # The overall system matrix is initialised to the identity matrix
    M_sys = np.eye(2, dtype=complex)

    for i in range(len(n_layers)):
        # Local optical phase thickness (delta) accumulated within layer i
        delta = k0 * n_layers[i] * d_layers[i] * cos_layers[i]
        c, s = np.cos(delta), 1j * np.sin(delta)
        eta = eta_layers[i]

        # Building the 2x2 characteristic matrix of the local layer i
        M_i = np.array([
            [c, s / eta],
            [s * eta, c]
        ], dtype=complex)

        # Accumulation: M_sys = M_i * M_sys
        # The order of multiplication is crucial: we start from the substrate
        # (exit) towards the incident medium (input), hence right to left.
        M_sys = np.dot(M_i, M_sys)

    # ========================================================================
    # Step 5: Extracting the Fresnel amplitude coefficients
    # ========================================================================
    # The overall matrix is applied to the substrate boundary conditions
    # Matrix equation: [B, C]^T = M_sys * [1, eta_exit]^T
    B = M_sys[0,0]*1 + M_sys[0,1]*eta_exit
    C = M_sys[1,0]*1 + M_sys[1,1]*eta_exit

    # Complex reflection (r) and transmission (t) coefficients
```

```
    r = (eta_inc * B - C) / (eta_inc * B + C)
    t = (2.0 * eta_inc) / (eta_inc * B + C)

    # ================================================================
    # Step 6: Power balance (energy conservation)
    # ================================================================
    # Reflectance is measured in the same medium, hence a plain squared
modulus
    R = abs(r)**2

    # Poynting-vector correction (energy flux):
    # The transmittance requires comparing the real parts of the admittances
    # of the outer media, because the wave changes propagation medium.
    T_factor = (eta_exit.real) / (eta_inc.real) if eta_inc.real > 1e-12 else
0.0
    T = T_factor * abs(t)**2

    # The missing energy is necessarily absorbed by the stack
    A = 1.0 - R - T

    return R, T, A

# --- Simple usage example (bare glass) ---
# For glass of index 1.5 in air (1.0) at 550 nm under normal incidence:
# R, T, A = compute_tmm(1.0, 1.5, [], [], 550)
# print(f"Reflectance of bare glass: {R*100:.2f}%") # Prints ~4.00%
```

Teaching document (Master's level, M2) — Based on H.A. Macleod, Thin-Film Optical Filters.

Online version and interactive simulator: **tpcouchesminces.vercel.app**